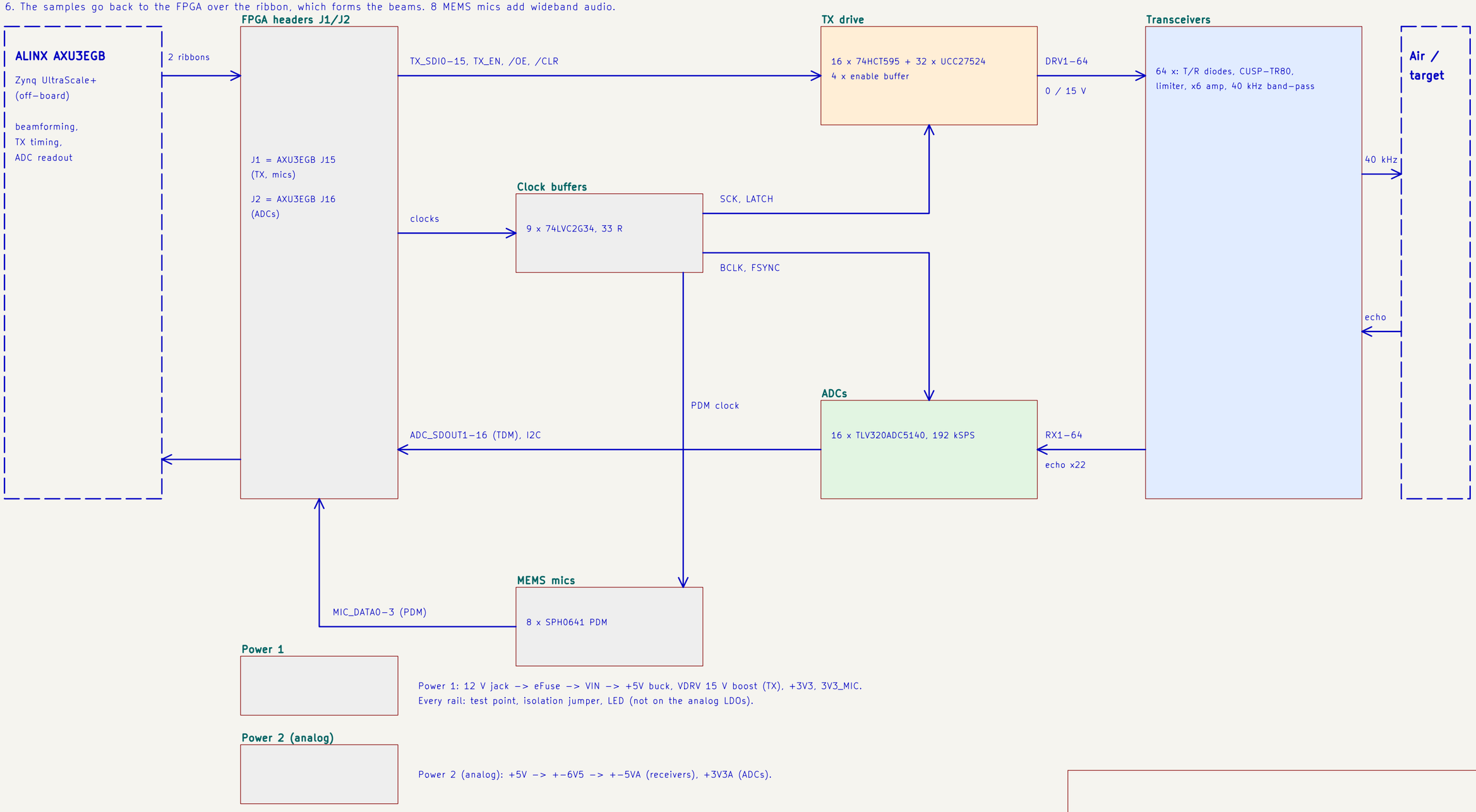
Hybrid 8x8 ultrasonic AESA: how the signal flows

1. The FPGA streams one TX bit per element (16 serial lines, 10 MHz) and the clocks.
2. TX drive: shift registers latch the bits; gate drivers switch each element between 0 and 15 V.
3. Transceivers: the 64 CUSP-TR80 elements emit 40 kHz; echoes return to the same elements.
4. The diode T/R switch and limiter protect the receiver; amplifier + 40 kHz band-pass give x22.
5. ADCs: 16 x TLV320ADC5140 sample all 64 channels at once, 192 kSPS (raw waveform, amplitude and phase).
6. The samples go back to the FPGA over the ribbon, which forms the beams. 8 MEMS mics add wideband audio.



Click a block to open it. Repeated parts are one sheet each: channel x64, adc x16, tx_group x16, en_buf x4, clk_buf x9, mic x8.

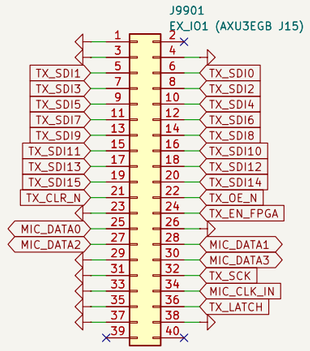
Hybrid 8x8 ultrasonic AESA	
Sheet: /	
File: aesa.kicad_sch	
Title: Hybrid 8x8 ultrasonic AESA: block diagram	
Size: A3	Date: 2026-10-20
KiCad E.D.A. 10.0.6	**Rev: 2**
	Id: 1/15

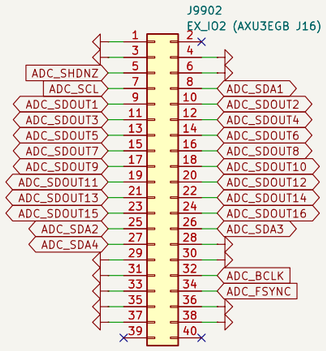
<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

AXU3EGB expansion headers (40-pin IDC, 2.54 mm). J1 = board J15 (TX, mics), J2 = board J16 (ADCs: 16 TDM lines, BCLK, FSYNC, I2C).

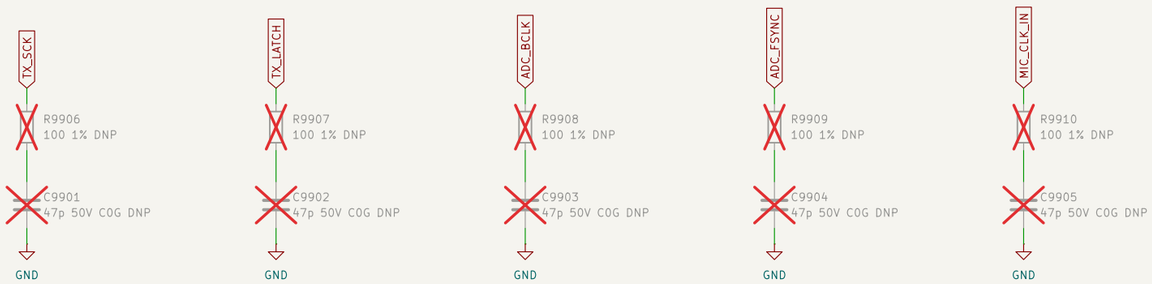
Pin 2 (+5V) and 39/40 (+3.3V) of the FPGA board are left open. Clocks sit between GND conductors (pins 31–37).

Spare FPGA pins tied to GND must stay unused (inputs) in the XDC.

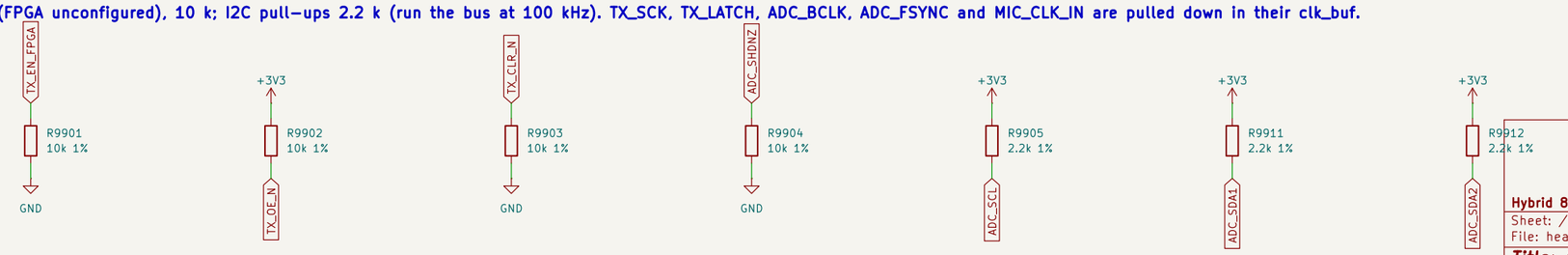




FPGA ribbon outputs: DRIVE 4, SLEW SLOW (near source-matched to the 100-ohm ribbon). DNP fallback: RC end termination 100 R + 47 pF per clock, fit only with DRIVE 8.



Idle pulls (FPGA unconfigured), 10 k; I2C pull-ups 2.2 k (run the bus at 100 kHz). TX_SCK, TX_LATCH, ADC_BCLK, ADC_FSYNC and MIC_CLK_IN are pulled down in their clk_buf.



Hybrid 8x8 ultrasonic AESA

Sheet: /FPGA headers J1/J2/

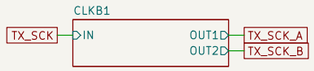
File: headers.kicad_sch

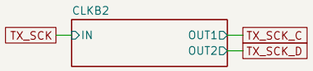
Title: AXU3EGB headers J15/J16, idle pulls

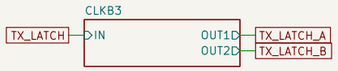
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 2/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

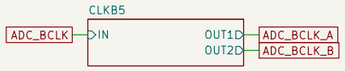
Clock buffers. Each FPGA clock enters one or two clk_buf cells; every output drives at most 4 loads through 33 R.

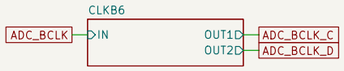


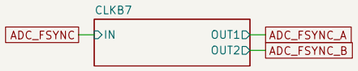


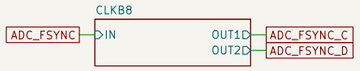














Hybrid 8x8 ultrasonic AESA

Sheet: /Clock buffers/
File: clocks.kicad_sch

Title: Clock buffers: 9 x 74LVC2G34 at the FPGA headers

Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 3/15

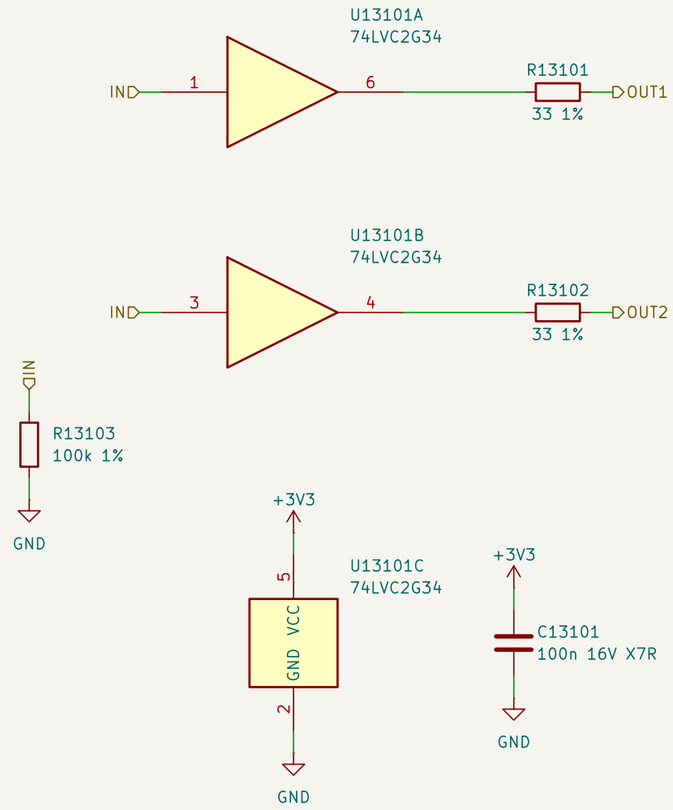
<< up to: Clock buffers: 9 x 74LVC2G34 at the FPGA headers

This sheet is used 9 times; the page shows the first instance.

Clock fan-out: 74LVC2G34 at the header, both gates on IN, 33 R series termination per output

Each output drives at most 4 loads (series termination works for a lumped load at the far end).

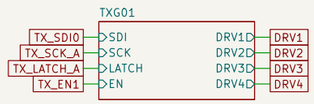
100 k pull-down keeps the clock low while the FPGA is unconfigured.

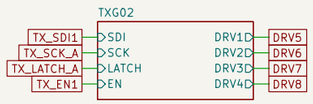


Hybrid 8x8 ultrasonic AESA		
Sheet: /Clock buffers/CLKB1/ File: clk_buf.kicad_sch		
Title: Clock buffer (x9): 74LVC2G34, 33 R outputs		
Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 4/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

TX drive. The FPGA streams 16 serial lines (TX_SDI0–15) into 16 tx_group cells; each drives 4 elements (DRV). TX_EN_FPGA reaches the drivers through 4 en_buf cells.





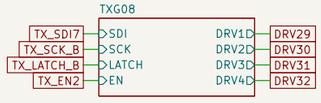


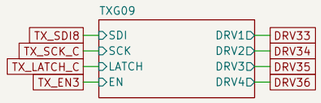




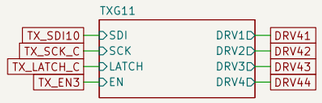


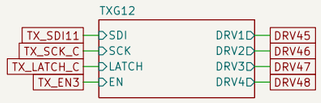














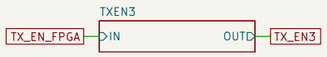














Hybrid 8x8 ultrasonic AESA

Sheet: /TX drive/
File: tx.kicad_sch

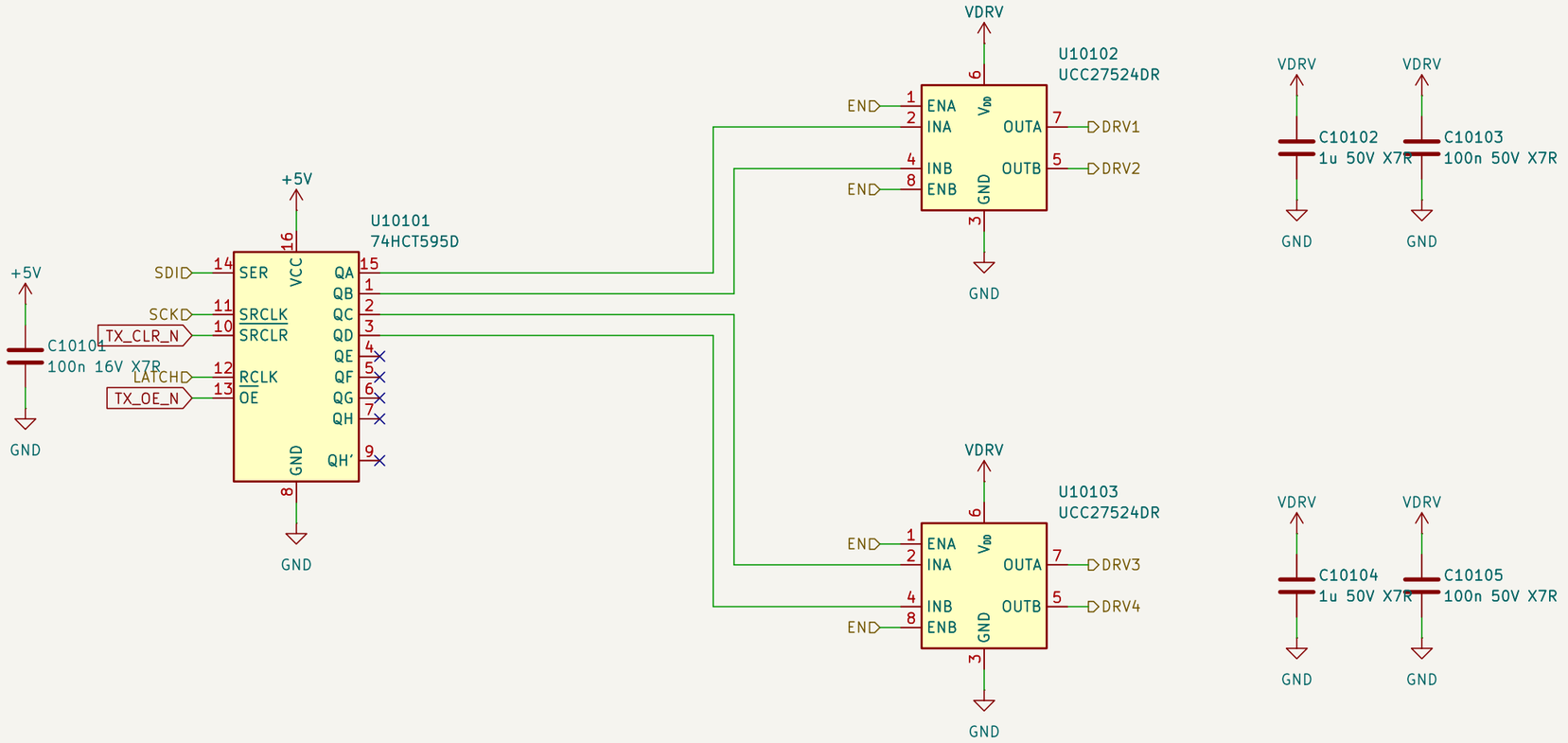
Title: TX drive: 16 shift–register/driver groups, 4 enable buffers

Size: A3	Date: 2026–10–20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 5/15

This sheet is used 16 times; the page shows the first instance.

TX group: 4 elements. 74HCT595 QA–QD -> UCC27524 inputs; QE–QH unused (16 data lines, 10 MHz SRCLK)

EN from one 74AHCT1G125 per 8 drivers (en_buf). SCK/LATCH from a clk_buf output (<= 4 loads).



Hybrid 8x8 ultrasonic AESA

Sheet: /TX drive/TXG01/

File: tx_group.kicad_sch

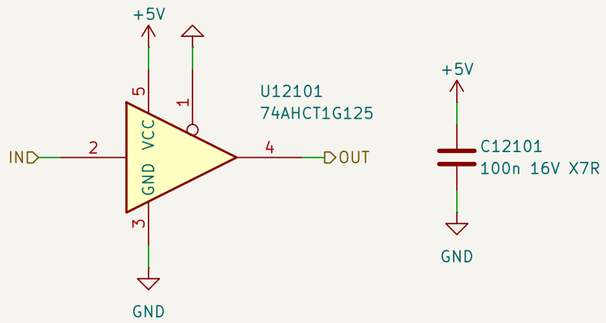
Title: TX group (x16): 74HCT595 + 2 x UCC27524, 4 elements

Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 6/15

<< up to: TX drive: 16 shift-register/driver groups, 4 enable buffers

This sheet is used 4 times; the page shows the first instance.

TX enable buffer: 74AHCT1G125, 3.3 V in (TTL threshold), 5 V out to 8 UCC27524 (16 EN pins)



Hybrid 8x8 ultrasonic AESA

Sheet: /TX drive/TXEN1/
File: en_buf.kicad_sch

Title: TX enable buffer (x4): 74AHCT1G125

Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 7/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

Transceiver array. Channel n = element n of the 8 x 8 grid, numbered down the columns (as on the board). Each channel takes DRVn from the TX drive and returns its amplified echo RXn to the ADCs.

CH01 DRV1 → DRV_OUT / RX_OUT → RX1	CH09 DRV9 → DRV_OUT / RX_OUT → RX9	CH17 DRV17 → DRV_OUT / RX_OUT → RX17	CH25 DRV25 → DRV_OUT / RX_OUT → RX25	CH33 DRV33 → DRV_OUT / RX_OUT → RX33	CH41 DRV41 → DRV_OUT / RX_OUT → RX41	CH49 DRV49 → DRV_OUT / RX_OUT → RX49	CH57 DRV57 → DRV_OUT / RX_OUT → RX57
CH02 DRV2 → DRV_OUT / RX_OUT → RX2	CH10 DRV10 → DRV_OUT / RX_OUT → RX10	CH18 DRV18 → DRV_OUT / RX_OUT → RX18	CH26 DRV26 → DRV_OUT / RX_OUT → RX26	CH34 DRV34 → DRV_OUT / RX_OUT → RX34	CH42 DRV42 → DRV_OUT / RX_OUT → RX42	CH50 DRV50 → DRV_OUT / RX_OUT → RX50	CH58 DRV58 → DRV_OUT / RX_OUT → RX58
CH03 DRV3 → DRV_OUT / RX_OUT → RX3	CH11 DRV11 → DRV_OUT / RX_OUT → RX11	CH19 DRV19 → DRV_OUT / RX_OUT → RX19	CH27 DRV27 → DRV_OUT / RX_OUT → RX27	CH35 DRV35 → DRV_OUT / RX_OUT → RX35	CH43 DRV43 → DRV_OUT / RX_OUT → RX43	CH51 DRV51 → DRV_OUT / RX_OUT → RX51	CH59 DRV59 → DRV_OUT / RX_OUT → RX59
CH04 DRV4 → DRV_OUT / RX_OUT → RX4	CH12 DRV12 → DRV_OUT / RX_OUT → RX12	CH20 DRV20 → DRV_OUT / RX_OUT → RX20	CH28 DRV28 → DRV_OUT / RX_OUT → RX28	CH36 DRV36 → DRV_OUT / RX_OUT → RX36	CH44 DRV44 → DRV_OUT / RX_OUT → RX44	CH52 DRV52 → DRV_OUT / RX_OUT → RX52	CH60 DRV60 → DRV_OUT / RX_OUT → RX60
CH05 DRV5 → DRV_OUT / RX_OUT → RX5	CH13 DRV13 → DRV_OUT / RX_OUT → RX13	CH21 DRV21 → DRV_OUT / RX_OUT → RX21	CH29 DRV29 → DRV_OUT / RX_OUT → RX29	CH37 DRV37 → DRV_OUT / RX_OUT → RX37	CH45 DRV45 → DRV_OUT / RX_OUT → RX45	CH53 DRV53 → DRV_OUT / RX_OUT → RX53	CH61 DRV61 → DRV_OUT / RX_OUT → RX61
CH06 DRV6 → DRV_OUT / RX_OUT → RX6	CH14 DRV14 → DRV_OUT / RX_OUT → RX14	CH22 DRV22 → DRV_OUT / RX_OUT → RX22	CH30 DRV30 → DRV_OUT / RX_OUT → RX30	CH38 DRV38 → DRV_OUT / RX_OUT → RX38	CH46 DRV46 → DRV_OUT / RX_OUT → RX46	CH54 DRV54 → DRV_OUT / RX_OUT → RX54	CH62 DRV62 → DRV_OUT / RX_OUT → RX62
CH07 DRV7 → DRV_OUT / RX_OUT → RX7	CH15 DRV15 → DRV_OUT / RX_OUT → RX15	CH23 DRV23 → DRV_OUT / RX_OUT → RX23	CH31 DRV31 → DRV_OUT / RX_OUT → RX31	CH39 DRV39 → DRV_OUT / RX_OUT → RX39	CH47 DRV47 → DRV_OUT / RX_OUT → RX47	CH55 DRV55 → DRV_OUT / RX_OUT → RX55	CH63 DRV63 → DRV_OUT / RX_OUT → RX63
CH08 DRV8 → DRV_OUT / RX_OUT → RX8	CH16 DRV16 → DRV_OUT / RX_OUT → RX16	CH24 DRV24 → DRV_OUT / RX_OUT → RX24	CH32 DRV32 → DRV_OUT / RX_OUT → RX32	CH40 DRV40 → DRV_OUT / RX_OUT → RX40	CH48 DRV48 → DRV_OUT / RX_OUT → RX48	CH56 DRV56 → DRV_OUT / RX_OUT → RX56	CH64 DRV64 → DRV_OUT / RX_OUT → RX64

Hybrid 8x8 ultrasonic AESA

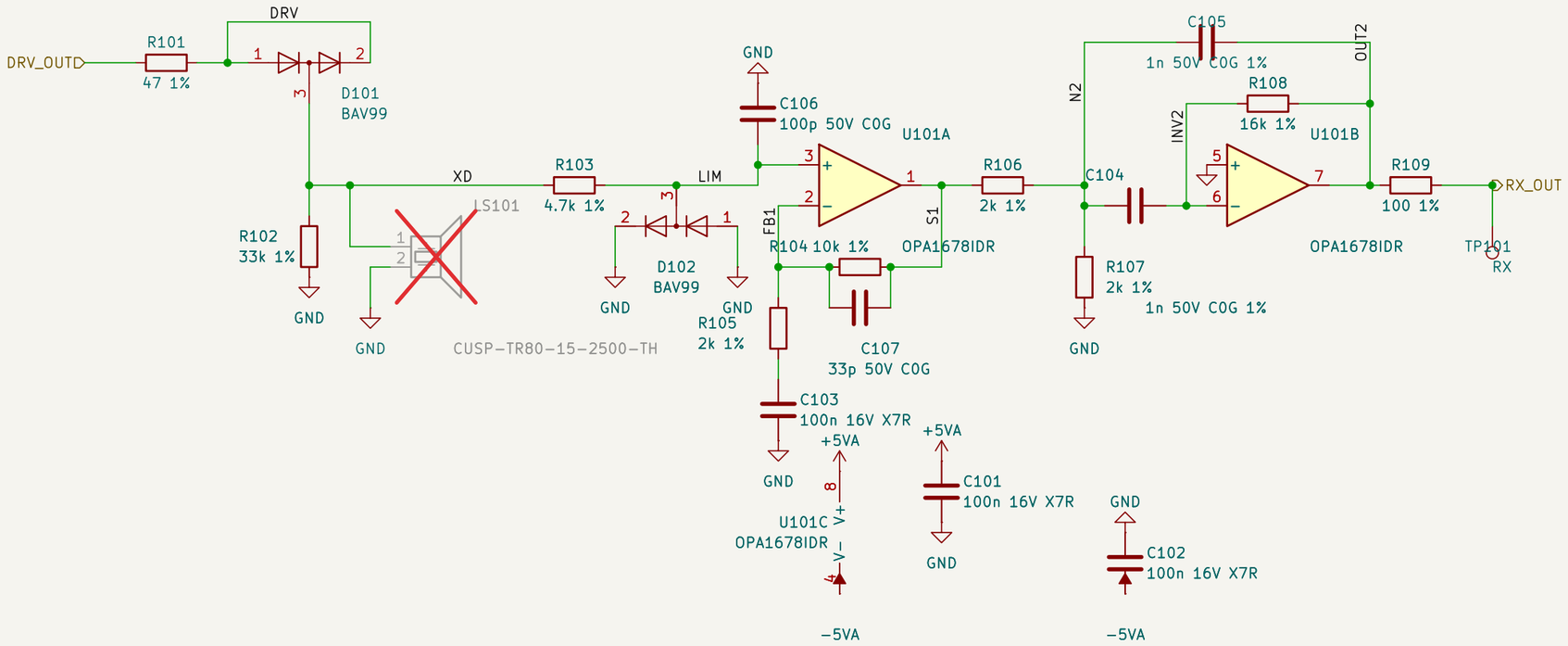
Sheet: /Transceivers/
File: array.kicad_sch

Title: Transceivers: 64 channels (8 x 8 grid)

Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 8/15

 This sheet is used 64 times; the page shows the first instance.

Transceiver channel (one of 64). TX: UCC27524 half -> 47 R -> BAV99 crossed diodes -> element (0–15 V square).
33k bleed resets the element after TX. RX: 4.7k + BAV99 clamp (+–0.7 V) + 100 pF (340 kHz RF filter).
OPA1678 stage 1: gain 6, AC-coupled at 800 Hz. Stage 2: MFB band-pass f0 39.8 kHz, Q 2, gain –4. Then 100 R to the ADC.
Element -> ADC +27 dB, pass band 31–51 kHz (sim/reports/rx_chain.md). Band-pass parts 1 % C0G for phase match.



Hybrid 8x8 ultrasonic AESA		
Sheet: /Transceivers/CH01/ File: channel.kicad_sch		
Title: Transceiver channel (x64): TX, T/R switch, RX amplifier		
Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 9/15

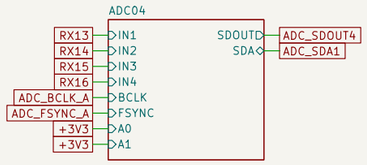
<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

ADCs. ADC k digitises RX 4(k−1)+1 … 4k and returns one TDM line (SDOUTk). Group g (ADC 4g−3 … 4g) shares SDAg and BCLK/FSYNC output g; A1/A0 give each ADC of a group its own I2C address.

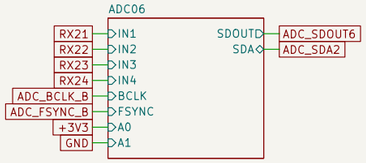


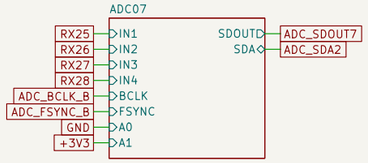


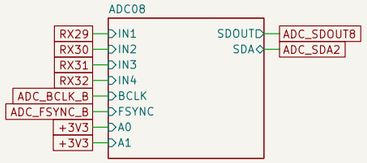














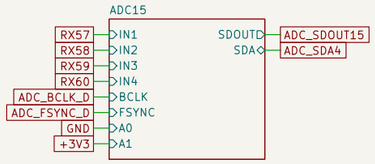


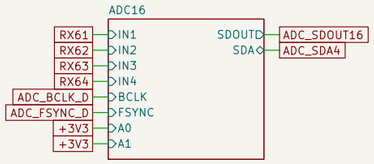












Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/
File: adcs.kicad_sch

Title: ADCs: 16 x TLV320ADC5140 (4 channels each)

Size: A3	Date: 2026−10−20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 10/15

ADC: TLV320ADC5140 (TI, 4 channels, up to 768 kSPS). Run at 192 kSPS: flat to about 86 kHz.

TDM slave: BCLK 18.432 MHz, FSYNC 192 kHz from the FPGA (shared by all 16 ADCs, so all 64 channels sample at the same instant). 4 slots x 24 bits on SDOUT.

I2C: SCL shared, SDA per group of 4 ADCs; A1/A0 set the address 0x4C–0x4F inside the group. Config writes can use I2C broadcast (P0_R2 I2C_BRDCAST_EN).

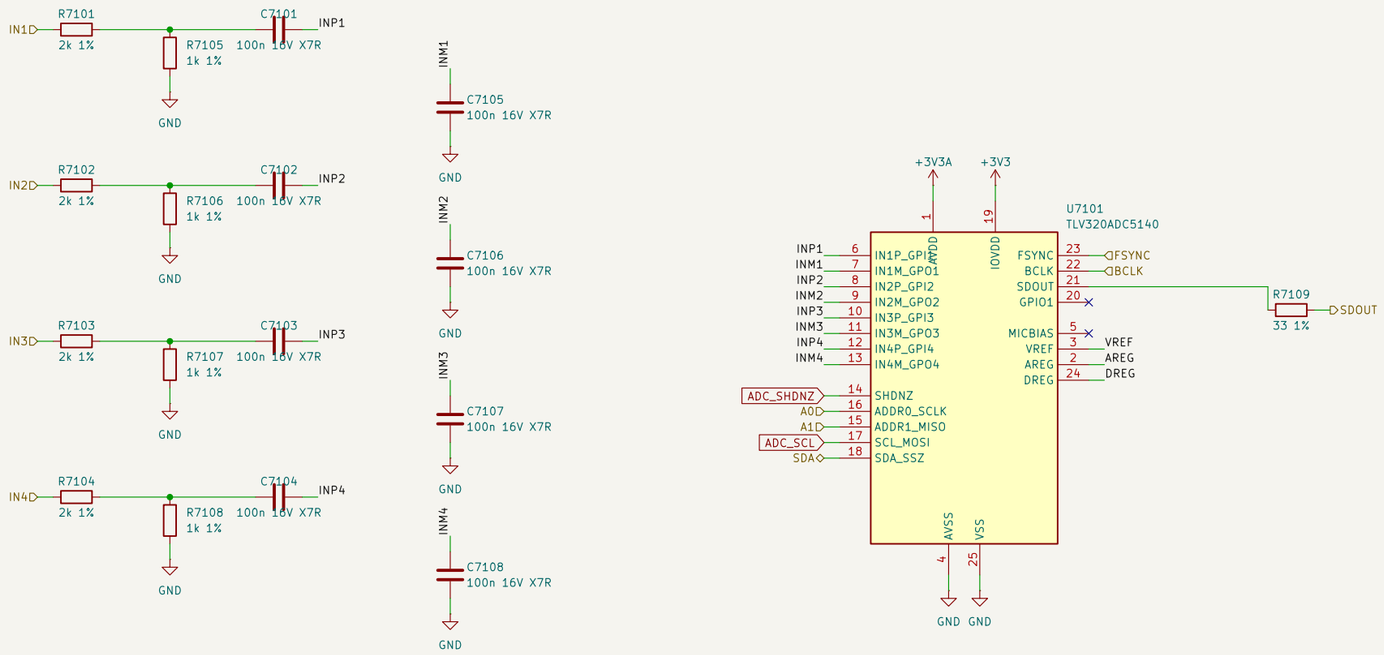
Per channel, set at boot: CHx_INSRC = single-ended (01), CHx_IMP = 10 k (01), AC-coupled, fixed 0 dB gain, DRE and AGC OFF (they change gain on the fly and break channel matching).

Also: ADC_FSCALE 2.75 V (1 Vrms single-ended full scale), AREG_SELECT = 1, TX_EDGE = 1 (the FPGA latches SDOUT one BCLK after launch).

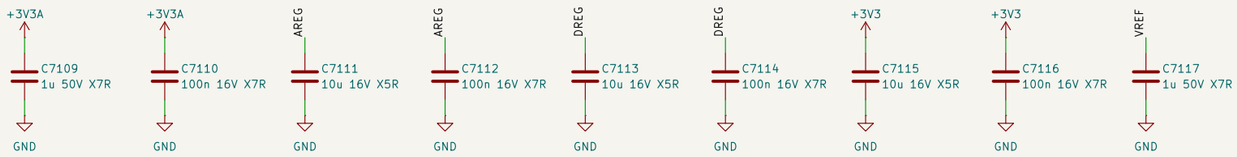
Input: 2k/1k divider (x0.31 with the 10 k input) keeps the receiver's +-4.2 V worst case inside the 1.41 V peak full scale; 100 nF couples to the ADC's own bias (150 Hz high-pass).

Power-up: SHDNZ (ADC_SHDNZ, pulled low on the header sheet) stays low until the rails settle, then wait 2 ms before I2C.

Channel inputs: 2k + 1k divider, 100 nF AC coupling (single-ended, INxM AC-grounded)



Decoupling at the pins (datasheet Figure 172): AVDD 1 uF + 100 nF, AREG and DREG 10 uF + 100 nF, IOVDD 10 uF + 100 nF, VREF 1 uF



Hybrid 8x8 ultrasonic AESA

Sheet: /ADCs/ADC01/

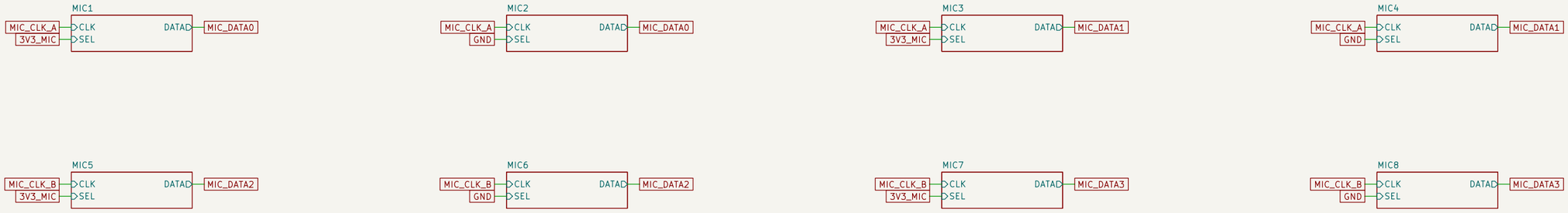
File: adc.kicad_sch

Title: ADC (x16): TLV320ADC5140, 4 channels, 192 kSPS TDM

Size: A3	Date: 2026-10-20	Rev: 2
KiCad E.D.A. 10.0.6		Id: 11/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

MEMS mics. Mics 2k−1 and 2k share MIC_DATAk−1 (SEL high on the odd one). MIC_CLK_A feeds mics 1−4, MIC_CLK_B mics 5−8.



Hybrid 8x8 ultrasonic AESA

Sheet: /MEMS mics/
File: mics.kicad_sch

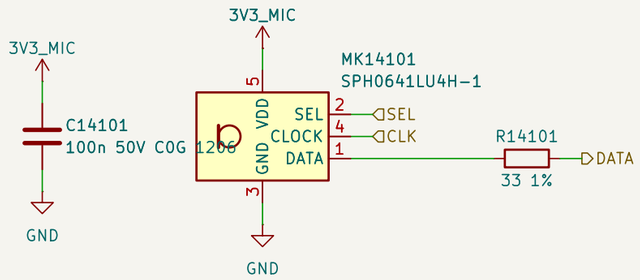
Title: MEMS microphones: 8 x SPH0641LU4H−1

Size: A3	Date: 2026−10−20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 12/15

<< up to: MEMS microphones: 8 x SPH0641LU4H-1 This sheet is used 8 times; the page shows the first instance.

MEMS mic SPH0641LU4H-1 (PDM). Two mics share DATA: SEL high on one, low on the other.

Mounted on the back, sound port through a board hole. 3V3_MIC is the ferrite-filtered rail (power sheet).

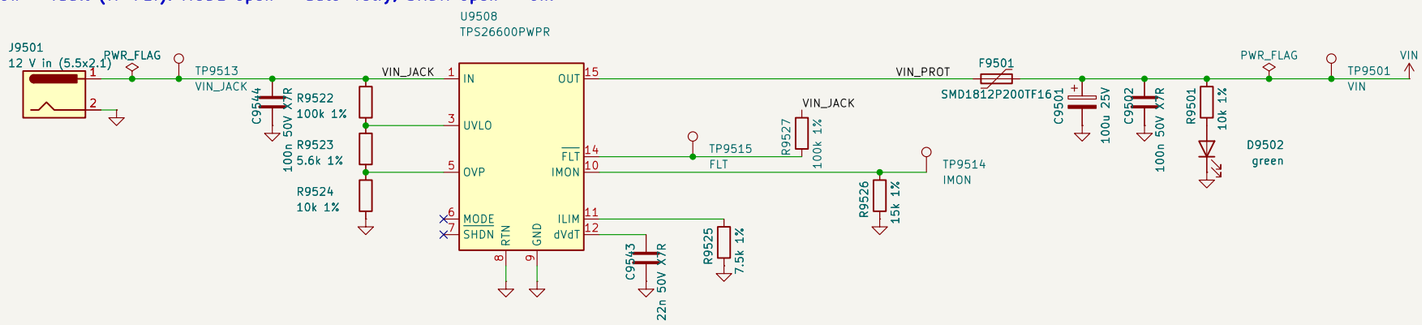


Hybrid 8x8 ultrasonic AESA		
Sheet: /MEMS mics/MIC1/		
File: mic.kicad_sch		
Title: MEMS mic (x8): SPH0641LU4H-1		
Size: A4	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 13/15

<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

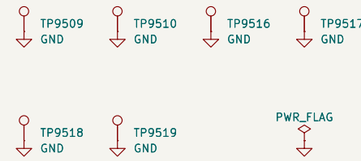
A. 12 V INPUT: jack -> TPS26600 eFuse -> polyfuse -> VIN

eFuse U8 (resettable, auto-retry): rated +-60 V (reverse polarity), on above 8.8 V, off above 13.8 V (protects the 14.5 V LM2731), current limit 1.6 A. F1 (2 A polyfuse) backs it up.
UVLO/OVP divider 100k / 5.6k / 10k. ILIM 12 k/A -> 7.5k = 1.6 A. dVdT 22 nF: 5 V/ms soft start. IMON 72 uA/A into 15k = 1.08 V/A (TP IMON). FLT low = fault (TP FLT). MODE open = auto-retry, SHDN open = on.



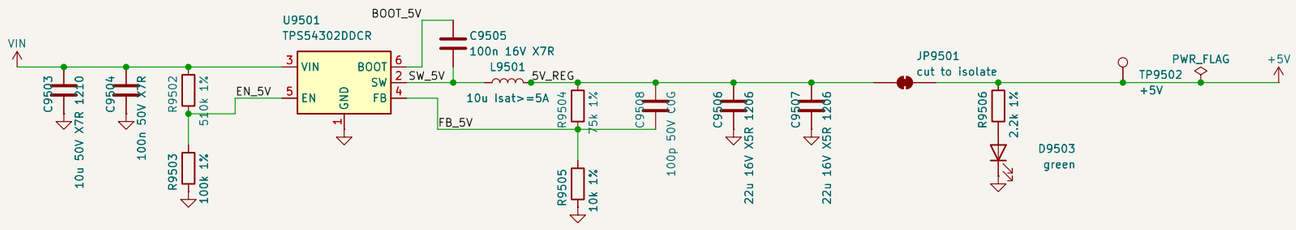
G. GROUND

One GND net and plane (In1, In2). Scope-ground test points: TP9/TP10 at the power supply, TP16-TP19 spread over the board (ADC row, array sides, TX row). Power tree and bring-up checklist: Power 2.



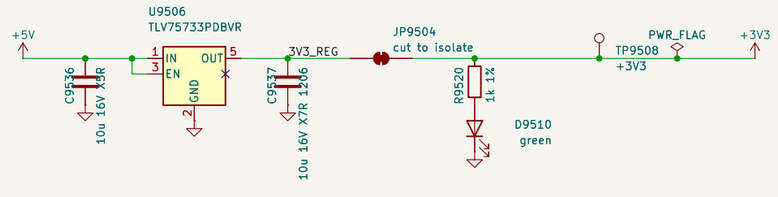
B. +5V: TPS54302 buck (TI reference design), 3 A

Vout = 0.596 V x (1 + 75k/10k) = 5.07 V. EN UVLO about 7.3 V. Load about 1.6 A. SW_5V loop (C3, U1, L1) kept tight on the board.



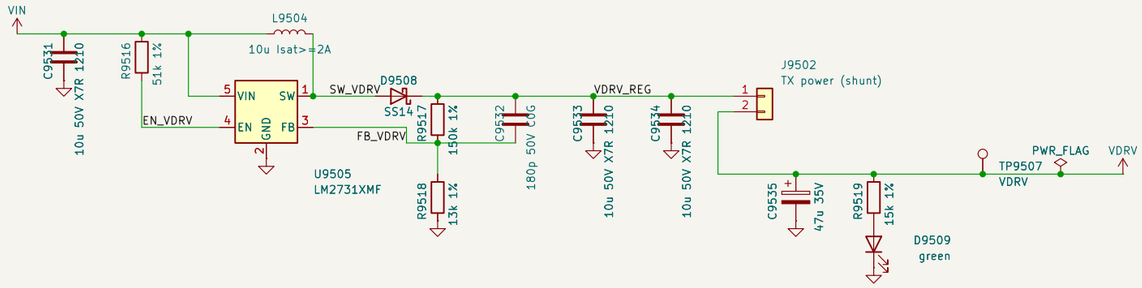
F. +3V3: TLV75733P LDO from +5V

1 A, stable with ceramic caps. Load < 0.15 A: ADC IOVDD, clock buffers, logic, mics.



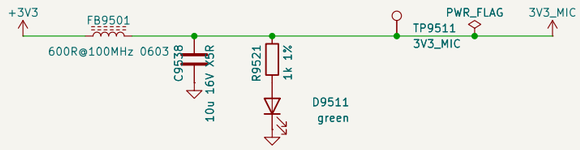
E. VDRV 15 V: LM2731X boost from VIN

1.23 V x (1 + 150k/13k) = 15.4 V. Load, all 64 elements in CW at 40 kHz: about 0.24 A (sim/reports/tx_cw.md). Remove the J2 shunt to unpower every TX driver.



H. 3V3_MIC: mic supply

Ferrite-filtered +3V3 for the 8 MEMS mics. Remove FB1 to isolate.

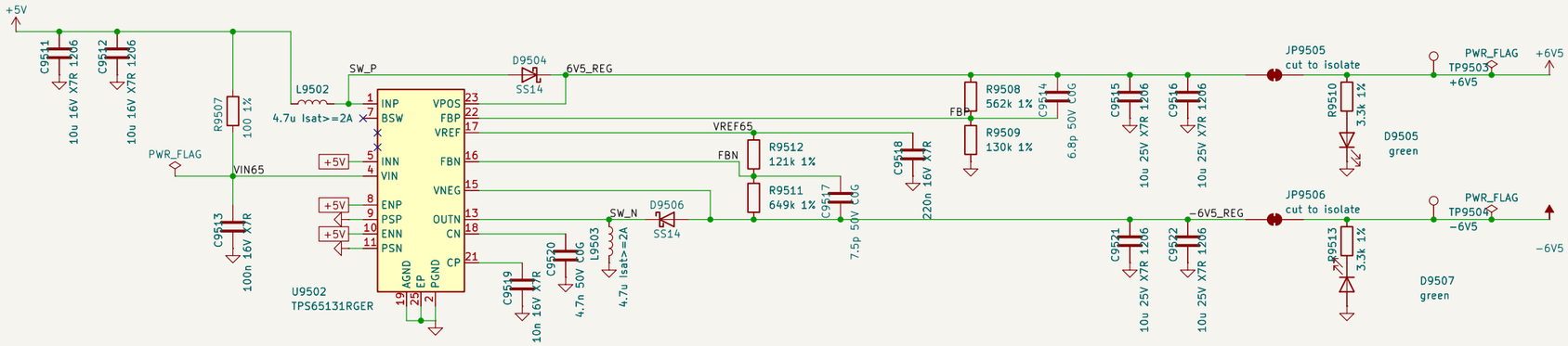


Hybrid 8x8 ultrasonic AESA		
Sheet: /Power 1/		
File: power.kicad_sch		
Title: Power 1: 12 V input, +5V, VDRV (TX), +3V3, 3V3_MIC		
Size: A3	Date: 2026-10-20	Rev: 2
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<< up to: Hybrid 8x8 ultrasonic AESA: block diagram

C. +−6V5: TPS65131RGER boost + inverter from +5V (feeds the analog LDOs only)

Drawn with the pin-identical TPS65130 symbol: order TPS65131RGER (2 A switches). PSP/PSN low: fixed-frequency PWM.
+6V5 = 1.213 V x (1 + 562k/130k) = 6.46 V. −6V5 = −1.213 V x 649k/121k = −6.51 V.

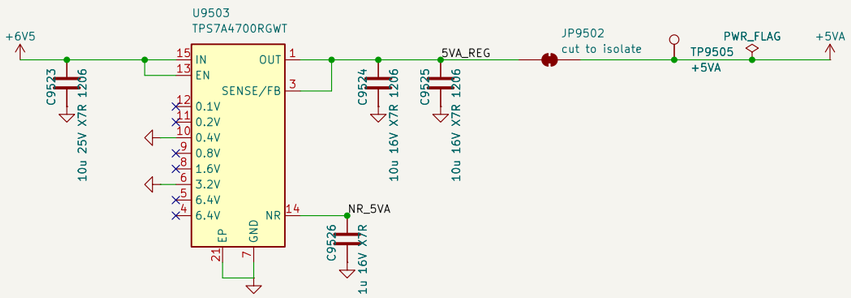


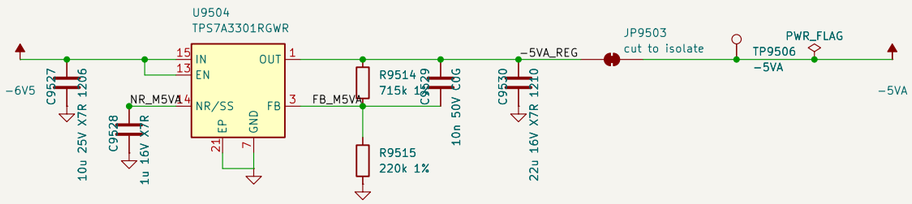
POWER TREE, BRING-UP

Power 1: A input, B +5V, E VDRV, F +3V3, H 3V3_MIC,
G ground. Power 2: C +−6V5, D +−5VA, I +3V3A.
12 V jack -> eFuse U8 -> F1 -> VIN.
VIN -> U1 buck -> +5V. VIN -> U5 boost -> VDRV.
+5V -> U2 -> +−6V5 -> U3 / U4 -> +−5VA (receivers).
+5V -> U7 -> +3V3A (ADC AVDD).
+5V -> U6 -> +3V3 -> FB1 -> 3V3_MIC.
First power-up: J2 shunt off, bench supply 12 V
limited to 0.3 A. Expect at the test points:
VIN 12 V, +5V 5.07 V, +3V3 3.30 V, 3V3_MIC 3.3 V,
+6V5 6.46 V, −6V5 −6.51 V, +5VA 5.00 V,
−5VA −5.00 V, +3V3A 3.30 V.
IMON 1.08 V per A in. FLT high (low = eFuse fault).
Then fit J2: VDRV 15.4 V.
A wrong rail: cut its jumper, measure the regulator
side (*_REG) and the load side apart.

D. +−5VA: TPS7A4700 / TPS7A3301 low-noise LDOs

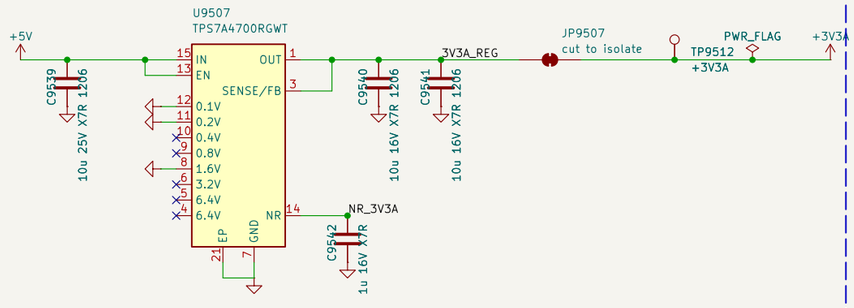
+5VA: ANY-OUT 1.4 + 3.2 + 0.4 = 5.0 V. −5VA = −1.176 V x (1 + 715k/220k) = −5.00 V.
Load: 64 x OPA1678 (4 mA per channel), about 0.3 A on each rail. Analog rails: no LEDs.





I. +3V3A: TPS7A4700 low-noise LDO from +5V

AVDD of the 16 ADCs (analog rail: no LED). ANY-OUT 1.4 + 1.6 + 0.2 + 0.1 = 3.3 V.
Load about 0.45 A; 0.8 W in the LDO, thermal vias under the pad.



Hybrid 8x8 ultrasonic AESA
Sheet: /Power 2 (analog)/
File: power_analog.kicad_sch
Title: Power 2 (analog): +−6V5, +−5VA, +3V3A
Size: A3 | Date: 2026-10-20 | **Rev: 2**
KiCad E.D.A. 10.0.6 | Id: 15/15